

# Notes: Agarwal, Lucca, Seru, and Trebbi (QJE, 2014)

## Inconsistent Regulators: Evidence from Banking

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# 1 Big picture

- **Question.** Does regulatory effectiveness depend only on written rules, or also on the institutions and incentives of the regulators who implement those rules?
- **Setting.** The paper studies U.S. state-chartered commercial banks, which are jointly overseen by state regulators and a federal regulator. The key regulatory output is the confidential CAMELS supervisory rating assigned during on-site examinations.
- **Core challenge.** Banks can often choose their charter and therefore partly choose their regulatory environment. That creates severe selection bias: lenient regulators may attract different banks, so raw comparisons across regulators are not causal.
- **Main identification idea.** The paper exploits legally determined *alternate examination policies* (AEPs), under which eligible state-chartered banks are examined by state and federal supervisors in an exogenously predetermined rotation. This allows the authors to compare the same bank under different regulators over time.
- **Main empirical result.** Federal regulators are systematically tougher than state regulators. In the baseline within-bank specification, federal examination spells raise a bank's composite CAMELS rating by about 0.095 notches relative to state examination spells. Since higher CAMELS means worse supervisory assessment, this is a meaningful tightening.
- **Downgrade-upgrade asymmetry.** Federal regulators downgrade banks much more often, while state regulators partially undo those moves by upgrading more frequently. In the pooled sample, about 62% of downgrades and 69% of harsh downgrades originate from federal regulators.
- **Operational response of banks.** Under federal oversight, banks report higher regulatory capital ratios, worse asset quality, lower return on assets, and higher expense ratios. Loan growth does not fall significantly.
- **Why this matters.** The evidence suggests that differences in *who* enforces the same rules can have real consequences. More lenient state supervision is associated with higher bank failure rates, more problem banks, lower repayment of TARP funds, and larger discounts when troubled bank assets are liquidated.
- **Main mechanism result.** The federal-state supervisory gap widens when local economic conditions are weak and narrows when state supervisory resources are stronger. The paper finds no support for a self-interest or revolving-door story as the main explanation.
- **Policy takeaway.** The institutional design of supervision matters independently of the formal rulebook. A dual regulatory system may create useful competition or local information advantages, but it can also generate delayed corrective action and costly inconsistency.

## 2 Data and measurement (key objects)

### 2.1 U.S. banking regulation and the alternating-exam setting

- U.S. commercial banks choose between a **national charter** and a **state charter**. Nationally chartered banks are supervised only by federal regulators, while state-chartered banks are jointly supervised by state banking departments and a federal agency.

- The relevant federal agency depends on Federal Reserve membership:
  - **State member banks (SMBs)** are supervised by the Federal Reserve and the state regulator.
  - **Nonmember banks (NMBs)** are supervised by the FDIC and the state regulator.
- The key supervisory tool is the **safety-and-soundness on-site examination**. It culminates in a **CAMELS** rating, a confidential supervisory score from 1 to 5, where lower values indicate fewer regulatory concerns.
- The six CAMELS subcomponents are capital adequacy, asset quality, management, earnings, liquidity, and sensitivity to market risk.
- CAMELS ratings are economically important because they affect deposit-insurance premia, access to the Fed’s discount window and government programs, and approval decisions involving licensing, branching, and mergers.

**Interpretation.** The paper needs a supervisory output that is comparable across regulators. CAMELS is exactly that object: it is a common rating scale applied by both federal and state examiners and directly tied to regulatory consequences.

## 2.2 Alternate examination policies (AEPs)

- The core institutional fact is that eligible state-chartered banks are assigned to **fixed rotations** between state and federal supervisors.
- These alternating policies were gradually developed in the 1970s and 1980s to reduce duplicative examinations and were standardized by federal legislation and regulatory guidance in the 1990s:
  - FDICIA (1991),
  - the Riegle Act (1994), and
  - FFIEC guidelines (1995).
- For banks under the AEP, on-site exams occur on legally determined **12-month** or **18-month** schedules depending on size and rating eligibility.
- Not every bank is eligible. Large banks, de novo banks, recently rechartered banks, and banks in weak condition are typically excluded.

**Interpretation.** The paper’s identification does not come from banks switching regulators by choice. It comes from a statutory rotation rule that changes the identity of the lead examiner for the same bank at predetermined intervals.

## 2.3 Sample construction and data sources

- **Main supervisory data.** The authors use examination records from the National Information Center of the Federal Reserve, including examiner identity and CAMELS ratings.
- **Bank outcomes.** They merge supervisory data with call-report measures of balance sheet composition, profitability, asset quality, and lending.

- **Regulatory resources.** They add budget and staffing information on state banking departments from the Conference of State Bank Supervisors (CSBS).
- **State-level outcomes.** They use state banking failure rates, problem-bank rates, TARP repayment information, and FDIC asset-sale discounts.
- **Sample window.** The main sample runs from 1996:Q1 to 2010:Q4, beginning shortly after the 1995 FFIEC harmonization of the alternating-exam framework.
- **AEP sample restrictions.** The sample is limited to state-chartered banks with assets below \$10 billion and with recent CAMELS ratings of 1 or 2, which are the banks most clearly covered by the alternating-exam rules.

## 2.4 Units of observation and key measured outcomes

- **Exam-level outcomes.** CAMELS ratings change only at the time of on-site examinations, so supervisory-rating regressions are run at the exam frequency.
- **Quarterly outcomes.** Bank balance-sheet and profitability variables evolve within a supervisory spell, so those regressions are run at the quarterly frequency.
- **Main bank-level outcomes.**
  - composite CAMELS and six subcomponents,
  - Tier 1 risk-based capital ratio,
  - leverage ratio,
  - return on assets (ROA),
  - efficiency ratio,
  - delinquency rate,
  - nonperforming loans to total loans,
  - loan growth.

## 2.5 Supervisory spells and why timing matters

- A **supervisory spell** is the period between an exam by one regulator and the next exam by the alternate regulator.
- CAMELS ratings are fixed within a supervisory spell, but quarterly bank outcomes can vary inside the spell.
- The paper compares the distribution of supervisory-rotation lengths with the distribution of exam-spell lengths and finds that the two coincide for roughly 95% of observations.

**Interpretation.** This matching between supervisory rotations and exam timing is important because it validates the claim that the sample is genuinely operating under the AEP rather than under ad hoc reassignments.

### 3 Models and empirical specifications (all equations + interpretation)

#### 3.1 Baseline regulatory-outcome equation

Let  $Y_{it}$  denote a supervisory outcome for bank  $i$  at time  $t$ , such as the composite CAMELS rating. The authors begin from

$$Y_{it} = \alpha + \mathbf{B}_{it}'\beta + \delta S_{it} + \mu_i + \lambda_t + \varepsilon_{it}, \quad (1)$$

where  $\mathbf{B}_{it}$  is a vector of bank characteristics,  $S_{it}$  is an indicator for a federal lead supervisor,  $\mu_i$  are bank fixed effects, and  $\lambda_t$  are quarter fixed effects.

After removing bank and time averages, the within-bank specification is

$$y_{it} = \mathbf{b}_{it}'\beta + \delta s_{it} + \varepsilon_{it}. \quad (1)$$

##### Interpretation.

- The coefficient  $\delta$  measures whether the same bank receives systematically different ratings when the lead supervisor is federal rather than state.
- Bank fixed effects absorb all time-invariant heterogeneity in charter, business model, and baseline risk.
- Quarter fixed effects absorb aggregate shocks and national regulatory environment changes.

#### 3.2 Why naive comparisons are biased: endogenous regulator choice

Absent the alternating-exam policy, regulator identity is endogenous. The authors formalize this with a reduced-form selection equation of the form

$$s_{it} = \pi y_{it} + \theta b_{it} + u_{it}, \quad (2)$$

where a bank's regulatory environment depends on expected treatment and bank characteristics.

##### Interpretation.

- This captures charter choice and regulatory shopping.
- A bank may pick the regulator most suitable to the activities it wishes to undertake.
- Then  $s_{it}$  is correlated with the error term in the ratings equation, making raw comparisons noncausal.

#### 3.3 Rotation policy as the source of exogenous supervisor assignment

For banks covered by the AEP, supervisor identity is instead governed by a policy rule:

$$s_{it} = p_{it} + \nu_{it}, \quad (3)$$

with the orthogonality condition

$$E[\nu_{it}\varepsilon_{it} \mid i \in \text{AEP}] = 0. \quad (4)$$

##### Interpretation.

- $p_{it}$  is the legally predetermined state-versus-federal rotation.
- Small deviations from the rotation policy can arise because of logistics or staffing, but the identifying assumption is that these deviations are random conditional on being in the AEP sample.
- This is what makes the within-bank comparison causal for eligible banks.

### 3.4 Observed bank behavior and reduced-form supervisory effects

The same policy can be used to study bank behavior itself. Let  $b_{it}$  denote a bank outcome such as capital, profitability, or delinquency. Then the authors write

$$b_{it} = \alpha_s s_{it} + v_{it}. \quad (5)$$

Substituting this into the within-bank ratings equation gives the reduced form

$$y_{it} = (\beta' \alpha_s + \delta) s_{it} + \varepsilon'_{it}. \quad (6)$$

#### Interpretation.

- The main coefficient on federal supervision captures both a *direct* effect on ratings and an *indirect* effect operating through observed and unobserved bank adjustments.
- When the authors add balance-sheet and profitability controls, they partially net out observed indirect channels.
- The fact that the controlled estimate is only slightly smaller than the baseline suggests that most of the supervisory effect operates directly through ratings, not through contemporaneous observables.

### 3.5 Exam-frequency rating regressions

The core CAMELS regressions use exam-level data and estimate whether a federal lead examiner raises the composite or component CAMELS score relative to a state examiner:

$$\text{CAMELS}_{it} = \alpha + \delta \text{FederalAgency}_{it} + \mu_i + \lambda_t + \varepsilon_{it}. \quad (2)$$

The paper also estimates versions that:

- separate the Fed from the FDIC,
- add bank-level controls,
- restrict to spells where rotation and exam length coincide exactly,
- and use fixed-effects IV as a robustness check with rotation rules and lagged supervisor identity.

**Interpretation.** These are all different ways to ask the same question: holding the bank fixed, is the federal supervisor tougher?

### 3.6 Quarterly bank-outcome regressions

For bank characteristics that can vary inside a spell, the specification is

$$X_{it} = \alpha + \rho \text{FederalAgency}_{it} + \mu_i + \lambda_t + u_{it}, \quad (3)$$

where  $X_{it}$  is a quarterly bank outcome such as capital ratio, ROA, efficiency, delinquency, or loan growth.

**Interpretation.**

- A positive coefficient on capital ratios and delinquency, together with a negative coefficient on ROA, is interpreted as evidence that federal regulators force more corrective action or stricter recognition of bank problems.
- Because banks may anticipate tougher federal exams and window-dress in advance, these estimates are likely a lower bound on the full difference in supervisory toughness.

### 3.7 State-level cost and consequence regressions

To assess whether leniency has real consequences, the paper augments the baseline CAMELS regression with interactions of the form

$$\text{CAMELS}_{it} = \alpha + \delta \text{FederalAgency}_{it} + \phi(\text{FederalAgency}_{it} \times Z_{st}) + \mu_i + \lambda_t + \varepsilon_{it}, \quad (4)$$

where  $Z_{st}$  is a state-level variable such as failure rates, problem-bank rates, TARP repayment, or FDIC asset-sale discounts.

**Interpretation.**

- A positive interaction means the federal–state spread is larger in states where that outcome is worse.
- This is not a pure causal design in the same sense as the baseline AEP comparison, but it is informative about whether lenient state supervision is correlated with costly outcomes.

### 3.8 Bank-level predictive-distress specification

The authors also ask whether past federal–state discrepancies predict future bank distress. A stylized version of their specification is

$$\Pr(\text{ProblemBank}_{it} = 1) = \alpha + \rho \overline{\Delta \text{CAMELS}}_{i,t-1} + \kappa \text{CAMELS}_{i,t-1} + \mu_i + \lambda_t + \varepsilon_{it}, \quad (5)$$

where  $\overline{\Delta \text{CAMELS}}_{i,t-1}$  is the lagged average difference between federal and state ratings for a given bank.

**Interpretation.**

- If regulatory inconsistency merely reflects different “styles” with no real consequences, past disagreement should not forecast future distress once past CAMELS is controlled for.
- Instead, the paper finds that larger past discrepancies predict higher future problem-bank probability even conditional on lagged CAMELS.



### 3.9 Mechanism regressions: local interests, weak regulators, self-interest

To understand why state regulators differ from federal regulators, the paper estimates interaction regressions using three families of explanatory variables:

1. **Local interests:** local unemployment and house prices,
2. **Weakness of regulators:** budget, number of examiners, training, experience, and examiner quality,
3. **Self-interest:** corruption, institutional quality, organizational structure, and revolving-door measures.

A generic specification is

$$\text{CAMELS}_{it} = \alpha + \delta \text{FederalAgency}_{it} + \phi(\text{FederalAgency}_{it} \times M_{st}) + \mu_i + \lambda_t + \varepsilon_{it}, \quad (6)$$

where  $M_{st}$  is one of the mechanism variables.

#### Interpretation.

- If local interests matter, federal–state differences should widen when the local economy is weak.
- If weak regulators matter, the gap should be smaller where state agencies have better staff and more training.
- If capture or self-interest matters, leniency should line up with corruption or revolving-door measures.

## 4 Identification and instruments (what makes the estimates credible)

### 4.1 Core identifying variation

- The main source of identification is the **exogenous alternation** of state and federal supervisors for banks under the AEP.
- This means the paper compares the **same bank** under different supervisors rather than comparing different banks regulated by different agencies.
- Bank fixed effects absorb all time-invariant heterogeneity, while quarter fixed effects absorb common shocks.

### 4.2 Why the within-bank design is powerful

- It eliminates the most important source of bias in this literature: endogenous sorting of banks into different regulatory settings.
- It also avoids the problem that different regulators may oversee systematically different portfolios of institutions.
- Since the paper focuses on CAMELS, a common supervisory metric, it compares directly comparable outcomes rather than trying to compare unrelated dimensions of supervisory activity.

### 4.3 Why this is not a conventional instrument paper

- The baseline design is not an external-instrument setup in the usual sense. Identification comes from institutional assignment rather than from an excluded instrument.
- That said, the authors do run **fixed-effects IV robustness checks**, instrumenting regulator identity with the AEP rule and lagged supervisor identity.
- They also restrict the sample to observations where supervisory rotation and exam timing coincide exactly and to exact four- or six-quarter spells. The results are nearly unchanged.

### 4.4 What is causally identified and what is not

- The paper identifies the average reduced-form effect of **federal versus state supervisor identity** on ratings and bank outcomes for eligible AEP banks.
- It does *not* separately identify whether federal regulators are too tough or state regulators are too lenient from the baseline design alone.
- It also does not separately identify all dynamic strategic interactions between regulators. The authors are explicit that the ratings estimates should be interpreted as reduced-form equilibrium effects.

### 4.5 Threats to identification and how the paper handles them

- **Deviations from exact rotation.** These exist because of scheduling or staffing, but the paper shows that they are rare and empirically unimportant.
- **Anticipation by banks.** Banks may prepare for tougher federal exams. This would attenuate estimated federal toughness, implying that the baseline effects are likely lower bounds.
- **Sample selection into AEP eligibility.** The paper deliberately studies relatively healthy, state-chartered banks because that is where exogenous alternation exists. This strengthens internal validity but narrows the direct scope of the causal estimates.

### 4.6 Main limitations

- The cleanest identification applies only to banks eligible for alternating examinations, not to the largest or weakest institutions.
- The mechanism and consequence regressions rely partly on cross-state heterogeneity and therefore do not have the same causal force as the within-bank AEP design.
- CAMELS captures a central supervisory judgment, but the paper cannot observe every informal supervisory action that accompanies the rating.

### 4.7 Relation to the literature

- The paper contributes to the literature on **regulatory design** by showing that identical formal rules can be implemented differently depending on institutional structure.

- It speaks to the literature on **regulatory shopping** and **competition among regulators** by showing how selection bias can arise and how to remove it.
- It also contributes to the empirical literature on **bank supervision** by using confidential supervisory ratings as a directly comparable measure of regulator behavior.

## 5 Tables and figures

### 5.1 Table I, Figure I, and Figure II: sample construction and timing

TABLE I  
SUMMARY STATISTICS OF STATE MEMBER BANKS AND NONSTATE MEMBER BANKS

|                         | Mean  | Std. dev. | Min    | Max    | Count |
|-------------------------|-------|-----------|--------|--------|-------|
| <b>NMBs</b>             |       |           |        |        |       |
| Tier1 RBCR              | 16.19 | 7.89      | 3.35   | 81.49  | 39150 |
| Leverage ratio          | 10.46 | 3.47      | 2.25   | 38.91  | 39150 |
| ROA                     | 1.02  | 0.91      | -7.08  | 5.97   | 39150 |
| Efficiency              | 65.60 | 17.74     | 0      | 210.49 | 39150 |
| Delinquency rate        | 2.69  | 2.43      | 0      | 22.35  | 39150 |
| Non performing to loans | 1.12  | 1.46      | 0      | 14.73  | 39150 |
| % Loan growth           | 2.63  | 6.11      | -29.19 | 47.14  | 39150 |
| CAMELS rating           | 1.65  | 0.62      | 1      | 5      | 39150 |
| <b>SMBs</b>             |       |           |        |        |       |
| Tier1 RBCR              | 15.90 | 7.76      | 3.64   | 81.24  | 5062  |
| Leverage ratio          | 10.29 | 3.26      | 2.62   | 35.23  | 5062  |
| ROA                     | 1.01  | 0.83      | -6.97  | 5.05   | 5062  |
| Efficiency              | 66.30 | 17.48     | 0      | 208.07 | 5062  |
| Delinquency rate        | 2.41  | 2.11      | 0      | 16.95  | 5062  |
| Non performing to loans | 1.00  | 1.26      | 0      | 14.93  | 5062  |
| % Loan growth           | 2.77  | 5.75      | -25.41 | 46.92  | 5062  |
| CAMELS rating           | 1.72  | 0.58      | 1      | 5      | 5062  |
| <b>All</b>              |       |           |        |        |       |
| Tier1 RBCR              | 16.16 | 7.87      | 3.35   | 81.49  | 44212 |
| Leverage ratio          | 10.44 | 3.45      | 2.25   | 38.91  | 44212 |
| ROA                     | 1.02  | 0.90      | -7.08  | 5.97   | 44212 |
| Efficiency              | 65.67 | 17.71     | 0      | 210.49 | 44212 |
| Delinquency rate        | 2.66  | 2.40      | 0      | 22.35  | 44212 |
| Non performing to loans | 1.10  | 1.44      | 0      | 14.93  | 44212 |
| % Loan growth           | 2.65  | 6.07      | -29.19 | 47.14  | 44212 |
| CAMELS rating           | 1.66  | 0.61      | 1      | 5      | 44212 |

Note: The table presents summary statistics for state chartered banks in our sample. NMBs are nonmember banks, SMBs are state member banks, and "all" include both SMBs and NMBs. The sample selection criteria are discussed in detail in Section II. The bank-level characteristics are: Tier 1 risk-based capital ratio, leverage ratio (Tier 1 capital as a share of total risk-weighted assets), efficiency ratio (noninterest expense as percent of net operating revenue), return on assets, share of nonperforming loans to total loans, and delinquency rate of the loan portfolio. Delinquent loans include loans that are 30-plus days past due and loans in nonaccrual status, and nonperforming loans that are 90-plus days delinquent and loans in nonaccrual status. Sample period is 1990Q1-2010Q4.

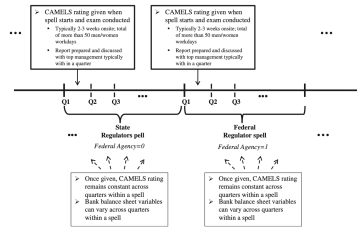


FIGURE I  
Supervisory Rotation Spells and Timing

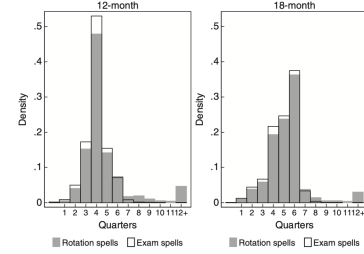


FIGURE II  
Distribution of Regulator Rotation and Examination Spells for SMBs and NMBs

Read:

- Table I reports descriptive statistics for the AEP sample of SMBs and NMBs. The average composite CAMELS is about 1.66, consistent with the paper's focus on relatively healthy banks that qualify for alternation.
- Figure I is a timing diagram. It clarifies the paper's key measurement distinction: CAMELS is assigned when the exam occurs and remains fixed until the next exam, whereas balance-sheet variables can move quarter by quarter inside a supervisory spell.
- Figure II validates the empirical design. The histogram of supervisory-rotation spells is almost identical to the histogram of exam spells, implying that supervisors do in fact alternate at the expected frequency for about 95% of the sample.

### 5.2 Table II and Table III: federal regulators are systematically tougher

Read:

- Table II is the core result. In the pooled sample of state-chartered banks, a federal lead supervisor raises the composite CAMELS rating by about 0.095 relative to a state supervisor in the same bank.
- The result is similar whether the federal regulator is the Fed or the FDIC and remains highly significant when the sample is restricted or when bank controls are added.

TABLE II  
IMPACT OF SUPERVISOR IDENTITY ON CAMELS RATINGS

|                            | (1)<br>Combined<br>CAMELS | (2)<br>Combined<br>CAMELS | (3)<br>Combined<br>CAMELS | (4)<br>Combined<br>CAMELS | (5)<br>Combined<br>CAMELS |
|----------------------------|---------------------------|---------------------------|---------------------------|---------------------------|---------------------------|
| Panel A: CAMELS regression |                           |                           |                           |                           |                           |
| Within-bank mean           | 1.751                     | 1.694                     | 1.700                     | 1.694                     | 1.699                     |
| Within-bank SD             | 0.303                     | 0.312                     | 0.311                     | 0.296                     | 0.318                     |
| FRB                        | 0.097***<br>[0.016]       |                           |                           |                           |                           |
| FDIC                       |                           | 0.095***<br>[0.012]       |                           |                           |                           |
| Federal agency             |                           |                           | 0.095***<br>[0.011]       | 0.081***<br>[0.009]       | 0.095***<br>[0.011]       |
| Other controls             |                           |                           |                           | Yes                       |                           |
| Constrained sample         |                           |                           |                           |                           | Yes                       |
| Cluster                    | State                     | State                     | State                     | State                     | State                     |
| Fixed effects              | Quarter<br>Bank ID        | Quarter<br>Bank ID        | Quarter<br>Bank ID        | Quarter<br>Bank ID        | Quarter<br>Bank ID        |
| Observations               | 5161                      | 39943                     | 45104                     | 44212                     | 41530                     |
| Adjusted R-squared         | 0.509                     | 0.470                     | 0.474                     | 0.570                     | 0.472                     |
| # of banks                 | 731                       | 5896                      | 6627                      | 6559                      | 6499                      |
| # of clusters              | 41                        | 50                        | 50                        | 49                        | 50                        |

|                         | (1)<br>Capital<br>rating | (2)<br>Asset<br>rating | (3)<br>Management<br>rating | (4)<br>Earnings<br>rating | (5)<br>Liquidity<br>rating | (6)<br>Sensitivity<br>rating |
|-------------------------|--------------------------|------------------------|-----------------------------|---------------------------|----------------------------|------------------------------|
| Panel B: Sub-components |                          |                        |                             |                           |                            |                              |
| Within-bank mean        | 1.501                    | 1.578                  | 1.809                       | 1.898                     | 1.573                      | 1.689                        |
| Within-bank SD          | 0.324                    | 0.373                  | 0.299                       | 0.330                     | 0.298                      | 0.274                        |
| Federal agency          | 0.074***<br>[0.010]      | 0.083***<br>[0.016]    | 0.119***<br>[0.010]         | 0.078***<br>[0.011]       | 0.057***<br>[0.009]        | 0.083***<br>[0.008]          |
| Cluster                 | State                    | State                  | State                       | State                     | State                      | State                        |
| Fixed effects           | Quarter<br>Bank ID       | Quarter<br>Bank ID     | Quarter<br>Bank ID          | Quarter<br>Bank ID        | Quarter<br>Bank ID         | Quarter<br>Bank ID           |
| Observations            | 45104                    | 45104                  | 45104                       | 45104                     | 45104                      | 39205                        |
| Adjusted R-squared      | 0.461                    | 0.402                  | 0.432                       | 0.491                     | 0.468                      | 0.405                        |
| # of banks              | 6627                     | 6627                   | 6627                        | 6627                      | 6627                       | 6309                         |
| # of clusters           | 50                       | 50                     | 50                          | 50                        | 50                         | 50                           |

Notes. The table reports results from an ordinary least squares regression that examines the effect of the federal regulator being the lead regulator in on-site examination on combined CAMELS rating (Panel A) and each subcomponent (Panel B). The sample selection criteria are discussed in Section II. All regressions include quarter and bank fixed effects and the standard errors are clustered at the state level. \*\*\* significant at 1% level. \*\* significant at 5% level. \* significant at 10% level. Sample period is 1996:Q1–2010:Q4.

TABLE III  
TABULATION OF COMPOSITE CAMELS UPGRADES AND DOWNGRADES

|                                                   | CAMELS<br>upgrade |           | CAMELS<br>downgrade |           | CAMELS<br>harsh downgrade |           |
|---------------------------------------------------|-------------------|-----------|---------------------|-----------|---------------------------|-----------|
|                                                   | Freq.             | Percent   | Freq.               | Percent   | Freq.                     | Percent   |
| SMBs, FRB-STATE rotating                          |                   |           |                     |           |                           |           |
| FRB                                               | 111               | 35        | 477                 | 73        | 199                       | 73        |
| STATE                                             | 208               | 65        | 178                 | 27        | 75                        | 27        |
| Total                                             | 319               | 100       | 655                 | 100       | 274                       | 100       |
|                                                   | Mean              | Std. dev. | Mean                | Std. dev. | Mean                      | Std. dev. |
| ΔCAMELS                                           | −1                | 0         | 1.09                | 0.33      | 1.22                      | 0.49      |
| NMBs, FDIC-STATE rotating                         |                   |           |                     |           |                           |           |
| FDIC                                              | 1222              | 46        | 3188                | 60        | 1687                      | 69        |
| STATE                                             | 1413              | 54        | 2103                | 40        | 769                       | 31        |
| Total                                             | 2635              | 100       | 5291                | 100       | 2456                      | 100       |
|                                                   | Mean              | Std. dev. | Mean                | Std. dev. | Mean                      | Std. dev. |
| ΔCAMELS                                           | −1                | 0         | 1.13                | 0.39      | 1.29                      | 0.54      |
| Federal Regulators-STATE rotating (SMBs and NMBs) |                   |           |                     |           |                           |           |
| Federal                                           | 1333              | 45        | 3665                | 62        | 1886                      | 69        |
| STATE                                             | 1621              | 55        | 2281                | 38        | 844                       | 31        |
| Total                                             | 2954              | 100       | 5946                | 100       | 2730                      | 100       |
|                                                   | Mean              | Std. dev. | Mean                | Std. dev. | Mean                      | Std. dev. |
| ΔCAMELS                                           | −1                | 0         | 1.13                | 0.39      | 1.28                      | 0.53      |

Notes. The table reports the frequency of composite CAMELS upgrades and downgrades by the lead supervisory agency (FRB, FDIC, or state) that took the regulatory action. Harsh downgrades are defined as downgrades such that the post-regulatory action CAMELS equals 3 or higher. The sample selection criteria are discussed in Section II. Sample period is 1996:Q1–2010:Q4.

- The management component shows the largest federal–state gap among the CAMELS subcomponents, which is exactly where supervisory discretion is likely to be greatest.
- Table III translates the regression result into frequencies. In the pooled sample, federal regulators account for about 62% of downgrades and 69% of harsh downgrades, while state regulators account for about 55% of upgrades.
- The right way to read these two tables together is: federal supervisors are not just slightly different on average; they are systematically more willing to move ratings downward.

### 5.3 Figure III: the sawtooth pattern in ratings

Read:

- Figure III plots the cumulative change in CAMELS ratings within a bank across future exam dates.
- The average path has a *sawtooth* shape: downgrades occur disproportionately during federal supervisory spells and are then partially offset by upgrades during subsequent state spells.
- This picture is especially useful because it makes clear that the federal–state gap is not just a one-time difference in levels. It repeatedly shows up whenever the lead supervisor rotates.

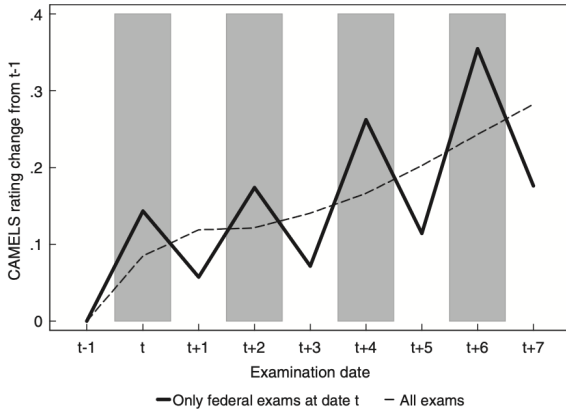


FIGURE III  
Average Evolution of CAMELS within a Bank

TABLE IV  
IMPACT OF SUPERVISOR IDENTITY ON BANK VARIABLES

|                       | (1)                | (2)                | (3)                 | (4)                    | (5)                      | (6)                 | (7)                |
|-----------------------|--------------------|--------------------|---------------------|------------------------|--------------------------|---------------------|--------------------|
|                       | Tier1<br>RBCR      | Leverage<br>ratio  | Efficiency<br>ratio | Return<br>on<br>assets | NPL to<br>total<br>loans | Delinquency<br>rate | Loan<br>growth     |
| Within-bank mean      | 16.091             | 10.436             | 67.077              | 0.956                  | 1.162                    | 2.657               | 2.959              |
| Within-bank SD        | 2.610              | 1.280              | 7.528               | 0.468                  | 0.845                    | 1.139               | 2.861              |
| Federal agency        | 0.066*             | 0.046*             | 0.326***            | -0.017***              | 0.038***                 | 0.044**             | -0.050             |
|                       | [0.038]            | [0.024]            | [0.093]             | [0.006]                | [0.011]                  | [0.017]             | [0.052]            |
| Cluster               | State              | State              | State               | State                  | State                    | State               | State              |
| Fixed effects         | Quarter<br>Bank ID | Quarter<br>Bank ID | Quarter<br>Bank ID  | Quarter<br>Bank ID     | Quarter<br>Bank ID       | Quarter<br>Bank ID  | Quarter<br>Bank ID |
| Observations          | 222836             | 223118             | 223215              | 222766                 | 222775                   | 223522              | 222137             |
| Adjusted<br>R-squared | 0.813              | 0.774              | 0.550               | 0.411                  | 0.374                    | 0.458               | 0.188              |
| # of banks            | 6610               | 6612               | 6620                | 6620                   | 6605                     | 6617                | 6596               |
| # of clusters         | 49                 | 49                 | 49                  | 49                     | 49                       | 49                  | 49                 |

Notes. The table reports results from an ordinary least squares regression that examines the effect of federal agencies being the lead regulator in on-site examination on banks' measures of asset quality, profitability, and loan growth. Column (1) looks at the Tier 1 risk-based capital ratio, column (2) looks at the leverage ratio, column (3) looks at expense (efficiency) ratio, column (4) looks at the ROA, column (5) looks at nonperforming loans share, column (6) looks at delinquency rates, and column (7) looks at new loan growth. The sample selection criteria are discussed in Section II. All regressions include quarter and bank fixed effects and the standard errors are clustered at the state level. \*\*\* significant at 1% level. \*\* significant at 5% level. \* significant at 10% level. Sample period 1996:Q1–2010:Q4.

### 5.4 Table IV: bank behavior under federal oversight

Read:

- Table IV asks whether banks behave differently when the lead supervisor is federal rather than state.
- Under federal oversight, banks report higher Tier 1 risk-based capital ratios and higher leverage ratios. This is consistent with stricter capital discipline.
- Federal spells are also associated with higher efficiency ratios, lower ROA, more nonperforming loans, and higher delinquency rates.

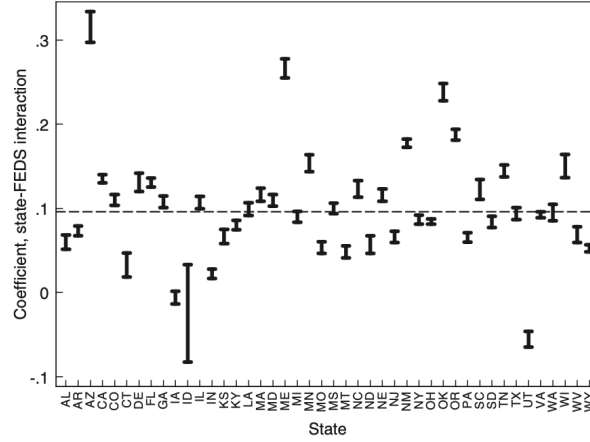


FIGURE IV  
Federal and State CAMELS Spread across States

- The pattern is exactly what one would expect if federal regulators force earlier recognition of problems and more costly corrective actions.
- Loan growth does not move significantly, so the evidence is not that federal regulators immediately cut off credit supply. The main action is in recognition, capitalization, and profitability.

## 5.5 Figure IV and Table V: heterogeneity and the costs of inconsistency

TABLE V  
ASSESSING COSTS AND BENEFITS OF INCONSISTENT REGULATION

|                                      | (1)                                                     | (2)                 | (3)                 | (4)                 | (5)                 |
|--------------------------------------|---------------------------------------------------------|---------------------|---------------------|---------------------|---------------------|
|                                      | Combined CAMELS                                         |                     |                     |                     |                     |
|                                      | Panel A: Costs and benefits using state level variables |                     |                     |                     |                     |
| Federal agency                       | 0.094***<br>[0.011]                                     | 0.094***<br>[0.010] | 0.093***<br>[0.010] | 0.095***<br>[0.010] | 0.092***<br>[0.009] |
| Federal agency * Failure rate        | 0.037***<br>[0.010]                                     |                     |                     | 0.001<br>[0.012]    |                     |
| Federal agency * Problem bank rate   |                                                         | 0.060***<br>[0.008] |                     |                     | 0.054***<br>[0.010] |
| Federal agency * TARP repayment      |                                                         |                     | -0.020*<br>[0.011]  |                     | -0.017*<br>[0.010]  |
| Federal agency * Asset sale discount |                                                         |                     |                     | 0.013**<br>[0.006]  | 0.016**<br>[0.007]  |
| Cluster                              | State                                                   | State               | State               | State               | State               |
| Fixed effects                        | Quarter                                                 | Quarter             | Quarter             | Quarter             | Quarter             |
|                                      | Bank ID                                                 | Bank ID             | Bank ID             | Bank ID             | Bank ID             |
| Observations                         | 45065                                                   | 45104               | 40823               | 44017               | 39999               |
| Adjusted R-squared                   | 0.556                                                   | 0.563               | 0.558               | 0.551               | 0.568               |
| # of banks                           | 6619                                                    | 6627                | 5995                | 6452                | 5866                |
| # of clusters                        | 49                                                      | 50                  | 41                  | 45                  | 39                  |

TABLE V  
(CONTINUED)

|                                  | (1)                                                    | (2)                 | (3)                 |
|----------------------------------|--------------------------------------------------------|---------------------|---------------------|
|                                  | Pr(Bank becomes a "Problem Bank" in a quarter)         |                     |                     |
|                                  | Panel B: Costs and benefits using bank level variables |                     |                     |
| Within-bank mean                 |                                                        | 0.028               |                     |
| Within-bank SD                   |                                                        | 0.128               |                     |
| Lagged mean difference           | 0.120***<br>[0.022]                                    | 0.063***<br>[0.014] | 0.057***<br>[0.013] |
| Lagged CAMELS                    |                                                        | 0.122***<br>[0.010] | 0.119***<br>[0.010] |
| Lagged rolling mean FED Abs Diff |                                                        |                     | 0.035***<br>[0.009] |
| Cluster                          | State                                                  | State               | State               |
| Fixed effects                    | Quarter                                                | Quarter             | Quarter             |
|                                  | Bank ID                                                | Bank ID             | Bank ID             |
| Observations                     | 160997                                                 | 160997              | 160997              |
| Adjusted R-squared               | 0.267                                                  | 0.401               | 0.401               |
| # of banks                       | 4992                                                   | 4992                | 4992                |
| # of clusters                    | 48                                                     | 48                  | 48                  |

Notes: This table reports results from an ordinary least squares regression that examines the effect of the federal regulators being the lead regulatory agency on CAMELS rating. Panel A assesses the costs and benefits of inconsistent regulation on state level variables that include bank failure rate, problem bank rate, TARP repayment rate, and asset sale discount. The level effects on interacted variables are included in all the regressions, but omitted in the table for brevity. Panel B presents bank level regressions that predict the propensity of a bank becoming a problem bank in a given regulatory spell as a function of lagged CAMELS federal-state spread at the bank level. All regressions include quarter and bank fixed effects and the standard errors are clustered at the state level. The sample selection criteria are discussed in Section II. \*\*\* significant at 1% level, \*\* significant at 5% level, \* significant at 10% level. Sample period 1986:Q1-2010:Q4.

Read:

- Figure IV shows that the federal-state spread is positive in most states but varies substantially in magnitude. So the main result is broad-based, not driven by one region.
- Table V, Panel A, interacts federal supervision with measures of state-level distress. The federal-state gap is larger in states with higher bank failure rates, higher problem-bank rates, lower TARP repayment, and larger FDIC asset-sale discounts.
- The economic message is that the states where local regulators are relatively more lenient are also the states where outcomes look worse.
- Table V, Panel B sharpens the argument at the bank level. Banks with larger past federal-state rating discrepancies are more likely to become problem banks in the future, even after controlling for lagged CAMELS.

- This is some of the paper’s strongest evidence that supervisory inconsistency is not harmless noise; it is associated with delayed intervention and future distress.

## 5.6 Table VI: why do differences exist between regulators?

|                                             | (1)                         | (2)<br>Combined CAMELS      | (3)                         | (4) |
|---------------------------------------------|-----------------------------|-----------------------------|-----------------------------|-----|
| <i>Panel A: Local interests</i>             |                             |                             |                             |     |
| Federal agency                              | 0.096***<br>[0.011]         | 0.096***<br>[0.011]         | 0.096***<br>[0.010]         |     |
| Federal agency * Local UR                   | 0.065***<br>[0.009]         |                             | 0.054***<br>[0.010]         |     |
| Federal agency * Local HPI                  |                             | −0.050***<br>[0.010]        | −0.020**<br>[0.008]         |     |
| Cluster<br>Fixed effects                    | State<br>Quarter<br>Bank ID | State<br>Quarter<br>Bank ID | State<br>Quarter<br>Bank ID |     |
| Observations                                | 45065                       | 45065                       | 45065                       |     |
| Adjusted R-squared                          | 0.483                       | 0.480                       | 0.485                       |     |
| # of banks                                  | 6619                        | 6619                        | 6619                        |     |
| # of clusters                               | 49                          | 49                          | 49                          |     |
| <i>Panel B: Weakness of regulators</i>      |                             |                             |                             |     |
| Federal agency                              | 0.096***<br>[0.011]         | 0.094***<br>[0.009]         | 0.095***<br>[0.010]         |     |
| Federal agency * Budget ratio               | −0.002<br>[0.004]           |                             | −0.003<br>[0.004]           |     |
| Federal agency * # Examiner ratio           | 0.007<br>[0.006]            |                             | 0.002<br>[0.004]            |     |
| Federal agency * Training ratio             | −0.010*<br>[0.006]          |                             | −0.007<br>[0.007]           |     |
| Federal agency * % Experienced examiner     | −0.013*<br>[0.007]          |                             | −0.015*<br>[0.008]          |     |
| Federal agency * Turnover                   |                             | −0.023*<br>[0.012]          | −0.026**<br>[0.011]         |     |
| Cluster<br>Fixed effects                    | State<br>Quarter<br>Bank ID | State<br>Quarter<br>Bank ID | State<br>Quarter<br>Bank ID |     |
| Observations                                | 43157                       | 43023                       | 41618                       |     |
| Adjusted R-squared                          | 0.474                       | 0.470                       | 0.470                       |     |
| # of banks                                  | 6354                        | 6299                        | 6098                        |     |
| # of clusters                               | 45                          | 40                          | 37                          |     |
| <i>Panel C: Self interest of regulators</i> |                             |                             |                             |     |
| Federal agency                              | 0.107***<br>[0.010]         | 0.087***<br>[0.026]         | 0.088***<br>[0.025]         |     |
| Federal agency * Corruption measure         | 0.002<br>[0.009]            |                             | −0.006<br>[0.008]           |     |
| Federal agency * Integrity rank             | 0.011<br>[0.010]            |                             | 0.005<br>[0.010]            |     |
| Federal agency * Institutional quality      | 0.010<br>[0.009]            |                             | 0.009<br>[0.009]            |     |
| Federal agency * Expenditure per capita     | 0.070***<br>[0.010]         |                             | 0.068***<br>[0.010]         |     |

|                                                                               | (1)                         | (2)<br>Combined CAMELS      | (3)                         | (4)                         |
|-------------------------------------------------------------------------------|-----------------------------|-----------------------------|-----------------------------|-----------------------------|
| Federal agency * Organization span                                            |                             | −0.014<br>[0.009]           | −0.014<br>[0.011]           |                             |
| Federal agency * Organization depth                                           |                             | 0.002<br>[0.004]            | 0.003<br>[0.003]            |                             |
| Federal agency * Turnover                                                     |                             | −0.024**<br>[0.010]         | −0.023*<br>[0.012]          |                             |
| Cluster<br>Fixed effects                                                      | State<br>Quarter<br>Bank ID | State<br>Quarter<br>Bank ID | State<br>Quarter<br>Bank ID |                             |
| Observations                                                                  | 40440                       | 41921                       | 37603                       |                             |
| Adjusted R-squared                                                            | 0.483                       | 0.470                       | 0.479                       |                             |
| # of banks                                                                    | 6333                        | 6151                        | 5884                        |                             |
| # of clusters                                                                 | 47                          | 39                          | 39                          |                             |
| <i>Panel D: Other bank level evidence and "Horse-Race" among explanations</i> |                             |                             |                             |                             |
| Within-bank mean                                                              |                             |                             |                             |                             |
| Within-bank SD                                                                |                             |                             |                             |                             |
| Federal agency                                                                | 0.094***<br>[0.011]         | 0.095***<br>[0.011]         | 0.094***<br>[0.011]         | 0.105***<br>[0.010]         |
| Federal agency * Size                                                         | 0.034***<br>[0.008]         |                             | 0.041***<br>[0.008]         | 0.013**<br>[0.006]          |
| Federal agency * Loans/assets                                                 | 0.000<br>[0.006]            |                             | 0.002<br>[0.006]            |                             |
| Federal agency * Public                                                       |                             | −0.010*<br>[0.005]          | −0.023***<br>[0.005]        | −0.016***<br>[0.005]        |
| Federal agency * Local UR                                                     |                             |                             |                             | 0.034***<br>[0.010]         |
| Federal agency * % Experienced examiner                                       |                             |                             |                             | 0.004<br>[0.009]            |
| Federal agency * Turnover                                                     |                             |                             |                             | −0.016<br>[0.012]           |
| Federal agency * Expenditure per capita                                       |                             |                             |                             | 0.047***<br>[0.012]         |
| Cluster<br>Fixed effects                                                      | State<br>Quarter<br>Bank ID | State<br>Quarter<br>Bank ID | State<br>Quarter<br>Bank ID | State<br>Quarter<br>Bank ID |
| Observations                                                                  | 45034                       | 45102                       | 45034                       | 38575                       |
| Adjusted R-squared                                                            | 0.479                       | 0.474                       | 0.479                       | 0.487                       |
| # of banks                                                                    | 6615                        | 6626                        | 6615                        | 6020                        |
| # of clusters                                                                 | 49                          | 50                          | 49                          | 40                          |

*Notes.* The table reports the results from an ordinary least squares regression that examines the effect of the federal regulator being the lead regulator in on-site examination on combined CAMELS rating. Panels A, B, and C test the "local interests," "weakness of regulators," and "self-interest of regulator" hypotheses outlined in the text. Panel D presents additional evidence using bank-level data as well as a horse race between the competing explanations. The level effects on interacted variables are included in all the regressions, but omitted in the table for brevity. We include quarter and bank fixed effects and the standard errors are clustered at the state level. \*\*\* significant at 1% level. \*\* significant at 5% level. \* significant at 10% level. Sample period 1996:Q1–2010:Q4.

Read:

- Table VI is the mechanism table. It organizes the evidence into three families of explanations: local interests, weak regulators, and self-interest.
- **Local interests.** The federal–state gap widens when local unemployment is high and when local economic conditions are weak. This supports the idea that state regulators are more reluctant to crack down when doing so would hurt local economies.
- **Weak regulators.** The gap narrows where state agencies devote more resources to training and where examiners are more experienced. This suggests that part of state leniency is explained by capacity constraints.
- **Self-interest.** Measures of corruption, institutional quality, and revolving doors do not explain the gap in a systematic way. This is why the paper rejects regulatory capture as the main story.

- The horse-race specifications point especially strongly toward local economic conditions as the dominant explanation.

## 5.7 Broader applicability and supplementary evidence

**Read:**

- The paper also studies broader samples that include national banks and thrifts. Once bank and time fixed effects are included, the main federal-toughness patterns remain similar, suggesting that time-invariant selection is a major part of charter-shopping bias.
- In additional evidence around the onset of the Riegle Act, the move from simultaneous supervision to alternating supervision makes state-chartered banks appear more leniently treated than under constant federal oversight.
- This supplementary evidence supports the paper’s broader claim: sharing oversight with more lenient local regulators can dilute discipline rather than creating an efficient good-cop/bad-cop arrangement.

## 6 Short summary

- **Method.** The paper uses a legally mandated rotation of state and federal bank supervisors to identify within-bank differences in supervisory behavior.
- **Main reduced-form result.** Federal supervisors are tougher: they downgrade more often, state supervisors upgrade more often, and the same bank receives worse CAMELS ratings under federal oversight.
- **Bank response.** Under federal supervisors, banks report higher capital ratios, worse asset quality, higher expenses, and lower profitability.
- **Consequences.** States with relatively lenient state regulators have worse banking outcomes, and past regulator disagreement predicts future distress.
- **Mechanism.** The main reason appears to be that state regulators place more weight on local economic conditions and, to a lesser extent, have weaker supervisory resources. The paper finds no evidence that capture or revolving doors are the main explanation.
- **Bottom line.** Regulatory institutions matter independently of formal rules. Identical prudential standards can be implemented inconsistently, and that inconsistency appears costly.

## 7 Conclusion

### 7.1 Core conclusion (what the paper establishes)

- The paper establishes that federal and state banking regulators apply the same supervisory framework differently.
- It also establishes that these differences are economically important rather than merely cosmetic: they show up in bank ratings, bank balance sheets, and subsequent distress outcomes.



- The paper’s central empirical contribution is to isolate these differences using exogenous within-bank regulatory rotation rather than cross-sectional comparisons contaminated by charter choice.

## **7.2 Economic intuition**

- State regulators appear to be softer because they weigh local economic conditions more heavily and may sometimes lack the financial or human resources needed for equally strict supervision.
- That softness can delay corrective actions. A bank that would be downgraded or pushed to recognize losses under a tougher federal regulator may instead receive more forbearance under a state regulator.
- Once one sees the regulatory system this way, it becomes natural that inconsistent oversight can create costly delay, opacity, and eventual larger losses.

## **7.3 Takeaway**

This paper is important because it shows that regulatory effectiveness is not pinned down by statutes alone. In banking, the same formal rules can generate different outcomes depending on who enforces them, how much those regulators care about local conditions, and how well resourced they are. The broader lesson is that optimal regulatory design must account not just for the content of rules, but for the incentives and capabilities of the institutions entrusted with implementing them.